

BSC 495: The Hidden World's Code

A Hands-On Fungal Genomics Lab

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Overview

The Mycological Curriculum for Education and Discovery (**Myco-Ed**) is a **Course-Based Undergraduate Research Experience (CURE)** that features team-based, hands-on experiments in fungal biology and genomics. You will learn new laboratory techniques and data analysis by starting with sample collection, fungal isolation, and the characterization of the resulting data to improve fungal genomic resources. Genomic DNA will be extracted from samples for genome sequencing, assembly, and annotation. By isolating and sequencing new fungal species, your team will help produce high-quality fungal reference genomes. This course will prepare you with workforce training in fungal biology and genomics and empower you to be a contributor to fungal genomics! Importantly, you will practice lab and bioinformatics skills that are broadly applicable!

To learn more about Quarto books visit [Quarto Books](#) To learn more about the MycoEd curriculum visit the [MycoEd GitHub](#))

Course Outcomes

- ** CO1**: Isolate fungal endophytes,
- ** CO2**: Characterize the morphology of fungal cultures through phenotypic and genomic assays,
- ** CO3**: Practice extracting fungal DNA,
- ** CO4**: Describe the process to PCR amplify fungal ITS,
- ** CO5**: Formulate meaningful and testable scientific questions/hypotheses,
- ** CO6**: Collect and critically interpret experimental data (including metadata),
- ** CO7**: Maintain an appropriate laboratory notebook,
- ** CO8**: Summarize their work and describe their findings in written and oral presentations.
- ** CO9**: Apply evidence-based strategies to improve scientific collaboration and team-work skills.
- ** CO10**: Reflect on your development as a scientist and team member through structured prompts and reflections.

Module Slides

The links below open the RevealJS slide decks for each module. Please note the slides can be advanced using the arrow keys on your keyboard, and you can also use the menu in the bottom right corner to navigate between slides. The slide decks are designed to be used in conjunction with the corresponding module content in this book, so feel free to refer back to them as needed while working through the modules.

Tip

Links to RevealJS slide decks for each module are listed below.

- [Open Module 0 RevealJS Slides](#)
- [Open Module 1 RevealJS Slides](#)
- [Open Module 2 RevealJS Slides](#)
- [Open Module 3 RevealJS Slides](#)
- [Open Module 4 RevealJS Slides](#)
- [Open Module 5 RevealJS Slides](#)
- [Open Module 6 RevealJS Slides](#)
- [Open Module 7 RevealJS Slides](#)

Introduction

This is a Quarto book created from the [MycoEd Workbook from March 2026](#) and adapted for the **BSC 495 *The Hidden World's Code***, a course developed for students at North Carolina State University. The course is designed to provide students with hands-on experience in fungal biology and genomics, and to empower them to contribute to the field of mycology through research and discovery.

The Mycological Curriculum for Education and Discovery (Myco-Ed) is a Course-Based Undergraduate Research Experience (CURE) that consists of coordinated hands-on experiments across fungal biology teaching labs with the goals of

- Training students and instructors in laboratory techniques and data analyses
- Compiling original data for improving fungal genomic resources
- Empowering students through isolation and genome sequencing of new fungal species

The content is organized into **modules**, each covering a specific topic related to mycology and education. The book is designed to provide comprehensive information and resources for educators, students, and anyone interested in learning about fungi and their role in the environment.

Module 0: Getting Started and Data Management

Module 0.1 – Lab Materials and Equipment Overview

This table lists all materials and equipment required across Myco-Ed modules. Use the search box to filter by module, item, company, or catalog number (web view), or scroll through the complete table below (in PDF and downloads).